Generative AI for Cybersecurity: LLM-Driven Attack Dataset Augmentation in Web API Detection

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***Abstract*—As Web APIs become integral to modern digital ecosystems, their increasing complexity and widespread adoption have made them appealing targets for cyberattacks. Effective detection and prevention mechanisms are crucial, and machine learning (ML) techniques provide promising solutions for identifying anomalous API traffic and addressing emerging threats. However, the effectiveness of ML-based security models is limited by the scarcity of realistic, up-to-date datasets and the rapid obsolescence of static training data in the face of evolving attack methodologies. To address these challenges, we present an approach that employs Large Language Models (LLMs) to generate synthetic Web API attack datasets. By leveraging LLMs, researchers can access abundant, customizable, and domain-specific synthetic data that closely simulate contemporary traffic patterns. This eliminates privacy and security concerns associated with real-world datasets and ensures a continuously adaptable data source for training and refining ML-based Intrusion Detection Systems. Ultimately, this strategy enables the development of more resilient ML models capable of effectively countering the dynamic and sophisticated nature of modern Web API threats.**

***Index Terms*—LLM, Generative AI, Data Augmentation, Web API, Intrusion Detection System, Privacy**

I. Introduction

Web APIs have become the cornerstone of modern digital ecosystems, facilitating seamless communication between diverse applications and systems [1]. However, their ubiquitous presence and critical role have also rendered them susceptible to cyberattacks. Threats such as injection vulnerabilities, broken object-level authorization, and sensitive data exposure, as detailed in the OWASP Top 10 API Security risks [2], highlight the growing sophistication and frequency of API-related security breaches. Consequently, the development of robust detection and prevention mechanisms has become a priority in the field of cybersecurity. Machine learning (ML) has emerged as a promising solution, offering capabilities to analyze network traffic, identify anomalies, and detect malicious activities [3], [4]. Nevertheless, the efficacy of ML models is contingent upon access to high-quality datasets that accurately reflect real-world traffic patterns and attack scenarios.

The primary challenge lies in the scarcity of realistic, diverse, and contemporary datasets for training and validating ML models. Privacy regulations, such as the General Data Protection Regulation (GDPR) [5], and the sensitive nature of network traffic data impose strict limitations on data sharing and usage. Organizations are often reluctant to expose their internal network data due to confidentiality breaches and potential exploitation risks. As a result, researchers are frequently confined to using publicly available datasets [6], [7], many of which need to be updated and capture the evolving nature of network behaviors and attack techniques. The static and limited scope of these datasets hinders the development of ML models capable of addressing the dynamic and sophisticated threats faced by modern Web APIs.

In addition to data scarcity, dataset obsolescence poses a significant issue. The rapid evolution of cyberattack methodologies makes static datasets insufficient for accurately representing the current threat landscape. Attack patterns that were prevalent a few years ago may no longer be relevant, while new threats are either not represented or missing from existing datasets. This disconnect highlights the need for a fundamental shift in generating datasets to ensure the continued relevance and effectiveness of machine learning-based security solutions.

To address these limitations, we investigate using LLMs to generate synthetic Web API attack datasets. LLMs, trained on extensive corpora of text and programming data, can produce realistic and customizable synthetic data that simulates contemporary traffic patterns and attack scenarios. This approach offers several advantages. Synthetic datasets generated by LLMs eliminate privacy and security concerns associated with real-world data, as they are not derived from actual network traffic. Moreover, LLMs allow for the tailoring of datasets to include specific attack vectors aligned with the latest trends in API security, thereby overcoming the static

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nature of traditional datasets.

II. Literature Review

A. Web API Security

Web API security encompasses the strategies and tools used to protect the data exchange interfaces between software systems. APIs have evolved into critical components of digital ecosystems—fueled by cloud computing, mobile services, and IoT—they have become prime targets for malicious activity. Exploited vulnerabilities can result in severe data breaches, service outages, and reputational damage [8], [9]. Evidence shows repeated API-related incidents, prompting guidance from entities like OWASP's API Security Top 10. By proactively securing APIs, organizations safeguard sensitive information, maintain operational stability, and uphold the trust that underpins today's interconnected digital environment.

B. Generative AI

Generative AI is a form of artificial intelligence designed to analyze existing data and create entirely new content in text, images, audio, and even video. By training on extensive datasets, these models discern underlying patterns, structures, and relationships, enabling them to produce novel outputs that resemble their training material's style, tone, or substance [10].

Unlike traditional AI methodologies that focus on classification, prediction, or pattern detection, generative AI takes a more proactive role in creativity. Instead of merely evaluating inputs, it synthesizes new information that may not have explicitly existed before. This capability opens the door to various innovative applications, including automated content creation, data augmentation, design prototyping, and interactive storytelling.

C. Large Language Model

Large Language Models have emerged as transformative tools that significantly advance how machines process, generate, and comprehend human language. Their introduction has reshaped numerous domains, enabling more natural interactions between humans and intelligent systems [11]. By learning statistical patterns from vast text data, LLMs can produce contextually coherent, semantically rich, and informative responses, often indistinguishable from human-generated text.

LLMs are built on a neural network architecture called the Transformer, which utilizes self-attention mechanisms. This architecture enables the model to assess the relevance of various words within a sentence in relation to one another. The model learns linguistic structures, grammar rules, and factual information by undergoing extensive pretraining on large datasets. After this pretraining phase, the model can be fine-tuned or directed to perform specific tasks, such as answering questions, summarizing texts, or generating code.

LLMs have demonstrated utility across diverse fields. In medicine, they assist with summarizing patient records, interpreting clinical guidelines, and supporting diagnostic reasoning [12]. In chemistry, they help analyze research literature, predict molecular structures, or generate hypotheses for new compounds [13]. As chatbots and digital assistants, LLMs facilitate natural, context-aware conversations for customer support, education, and entertainment. Beyond these domains, LLMs are increasingly integrated into software development, legal contract analysis, and content generation, reflecting their broad versatility.

LLMs are prime examples of foundation models, large-scale pre-trained models that serve as general-purpose building blocks for downstream tasks. Foundation models capture a wealth of linguistic and world knowledge, enabling developers and researchers to swiftly adapt them for specific applications with minimal additional data and training. This paradigm reduces the need to build models from scratch, cutting development time and computational costs.

One of the most well-known LLMs models is LLaMa (Large Language Model Meta AI) [14]. LLaMa variants, trained on extensive corpora of web text, can generate coherent narratives, answer questions, translate languages, and even reason about code.

D. Intrusion Detection Systems

Intrusion Detection Systems (IDS) are essential for enhancing API security by monitoring traffic and identifying malicious activities. Generally, IDS can be categorized into three main types: signature-based, anomaly-based, and hybrid-based detection [15]. Signature-based methods use predefined patterns of known attacks to identify threats in real time. While these systems effectively recognize familiar exploit patterns, they often struggle with new or disguised threats that do not match any known signatures.

Machine learning-based Intrusion Detection Systems utilize hybrid analysis to detect new or evolving threats aimed at Web APIs. However, their effectiveness is limited by a lack of sufficient data and the use of outdated datasets, which can lower detection accuracy and increase false positives. As attack patterns change rapidly, it is essential to continuously update and expand the training data to ensure strong, adaptive, and reliable API security.

III. Methodology

In this research, we present a model that leverages a LLMs to generate synthetic datasets. The complete methodology is illustrated in Figure 1.

A. HTTP API Dataset

The API Traffic Research Dataset Framework (ATRDF) [7] contains 127,500 API calls, encompassing HTTP request and response pairs. Each API call is structured with an HTTP request, including elements such as the HTTP method, headers, and payload as shown in Table I. Corresponding HTTP responses provide status codes and JSON message bodies that indicate the request's outcome. The dataset includes a mix of benign traffic and anomalous data representing various attack patterns.

The pipeline begins by loading the ATRDF HTTP API dataset. In this initial phase, raw traffic logs are parsed to

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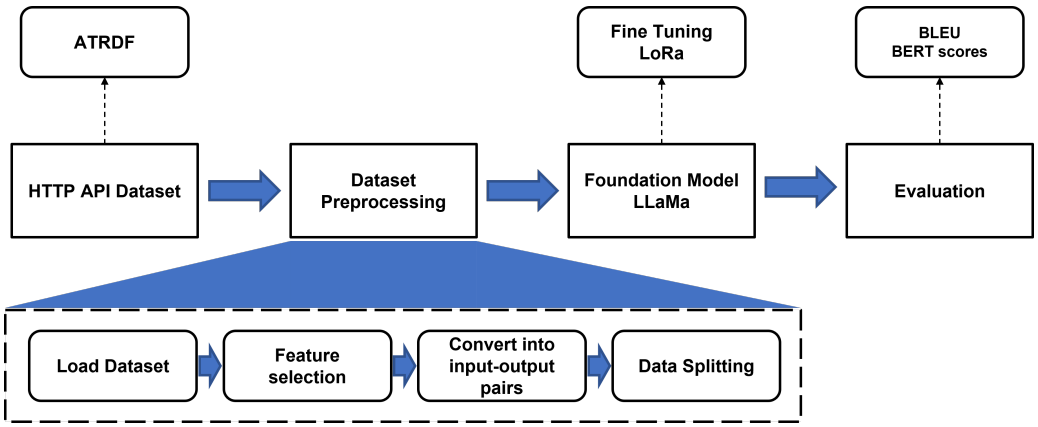


Fig. 1. Model Architecture.

TABLE I
EXAMPLE HTTP TRAFFIC OF API REQUESTS AND RESPONSES

Name	Example
request.headers.Host	127.0.0.1:5000
request.headers.User-Agent	Mozilla/5.0 (Windows NT 5.1; rv:31.0)...
request.headers.Accept-Encoding	gzip, deflate, br
request.headers.Accept	*/*
request.headers.Connection	keep-alive
request.headers.Accept-Language	de-CH
request.headers.Sec-Fetch-Site	none
request.headers.Sec-Fetch-Mode	same-origin
request.headers.Sec-Fetch-User	?1
request.headers.Sec-Fetch-Dest	document
request.headers.Set-Cookie	['ck=w6oM9XV_jvWjuue...
request.headers.Date	Fri, 11 Nov 2022 03:12:09 GMT
request.method	GET
request.url	http://127.0.0.1:5000/static/download_txt/7115
request.body	
response.status	404 NOT FOUND
response.headers.Content-Type	application/json
response.headers.Content-Length	32
response.status_code	404
response.body	"error": "File not found"
response.headers.Location	NaN
request.Attack_Tag	NaN
request.headers.Cookie	NaN
request.headers.Content-Length	NaN
response.headers.Set-Cookie	NaN

extract structured features such as request methods, headers, payloads, response codes, and messages. Any malformed data entries are filtered out to ensure quality and consistency, resulting in a clean, reliable dataset suitable for subsequent processing.

B. Dataset Preprocessing

Once the dataset has been cleaned, feature selection is applied to isolate the most informative elements of each HTTP transaction. This process involves identifying which headers, payload fields, status codes, or response message elements are most indicative of underlying patterns—be they benign traffic or malicious behaviors. Focusing on key features clarifies the dataset and reduces noise, improving fine-tuning effectiveness. The selected features are then encoded into a canonical text-based format that LLMs can easily process.

A critical transformation follows the feature selection: converting the refined HTTP API dataset into structured input-output pairs as shown in figure 2. Each pair presents a well-defined prompt and a corresponding reference response. For example, the prompt might ask the model to generate a "Benign" HTTP request while the output provides the correctly formatted HTTP request/response structure. Similarly, the prompt specifies an attack scenario for anomalous cases, and the output includes an appropriate, labeled response. This pairing ensures the data aligns with the model's training objectives, guiding the LLM to learn domain-relevant relationships.

With the input-output pairs established, the dataset is partitioned into training, validation, and testing subsets. The training set is the largest portion, enabling the model to learn patterns and associations thoroughly. The validation set is used to tune hyperparameters and avoid overfitting, while the test set provides an unbiased assessment of the model's final performance. This strategic data splitting ensures the model's learned capabilities generalize well beyond the training domain.

C. Fine Tuning Foundation Model

The input-output pairs are then used to fine-tune a foundation model (FM)—a pre-trained LLM known for its language understanding capabilities. This step tailors the FM to the specific nuances of HTTP API traffic. Techniques such as parameter-efficient fine-tuning (e.g., LoRA [16]) are employed to adapt the FM with minimal computational overhead and without updating the entire parameter space. The resulting specialized model can accurately generate requests, parse re-

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```
"input":
  "Generate an HTTP request with label: Benign",
"output": "{
  \"request.headers.Host\": \"127.0.0.1:5000\",
  \"request.headers.User-Agent\": \"Mozilla/5.0\",
    ...
  \"label\": \"Benign\"
}"
```

Fig. 2. Input-Output Pairs.

sponses, and identify anomalies in HTTP API traffic patterns, reflecting the knowledge distilled from the ATRDF dataset.

D. Evaluation criteria

Once fine-tuned, the model's performance is evaluated against the test set using established metrics. We evaluated the model using widely recognized benchmark metrics for language models, including BLEU (Bilingual Evaluation Understudy) and BERT (Bidirectional Encoder Representations from Transformers) scores. These automated metrics objectively assess how similar the model's generated responses are to reference answers in the test set. Importantly, they are highly correlated with human quality evaluations. BLEU, one of the oldest and most established metrics, examines word-level overlaps between the generated responses and the reference answers. In contrast, the BERT score evaluates semantic similarity by comparing the sentence representations of the generated outputs with those of the reference answers. The scores range from 0 to 1, with a score of 1 indicating perfect alignment. Together, these metrics provide a robust and comprehensive evaluation of the model's performance.

IV. Conclusion

In this paper, we introduce a model that leverages LLMs to generate synthetic datasets, directly addressing the twin challenges of data scarcity and obsolescence. By providing abundant, domain-specific data on demand, this approach fosters a more resilient development environment for machine learning models. As a result, these models can be continuously refined, validated, and enhanced, ensuring their robustness against the evolving threats targeting Web API security.

In future work, we plan to integrate Retrieval-Augmented Generation (RAG) techniques to mitigate the hallucination issues that often arise in LLMs. By incorporating external knowledge sources and reference materials, RAG can maintain the factual accuracy, consistency, and timeliness of the information used by the models. Additionally, RAG helps mitigate potential biases by dynamically retrieving diverse and authoritative sources of information, reducing reliance on potentially skewed or incomplete training data. By grounding responses in verifiable facts rather than purely learned patterns, RAG minimizes the risk of propagating biases inherent in the model's pretraining data. This refinement is expected to produce more reliable synthetic data and inform more effective decision-making, ultimately strengthening the overall security posture in an ever-changing cyber landscape.

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